HDOC DAY-55

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BATCH-15

Question

Write a C program using a variable-length array and pointers to find the maximum element in every sliding window of size k using nested loops. Print -1 if k is invalid.

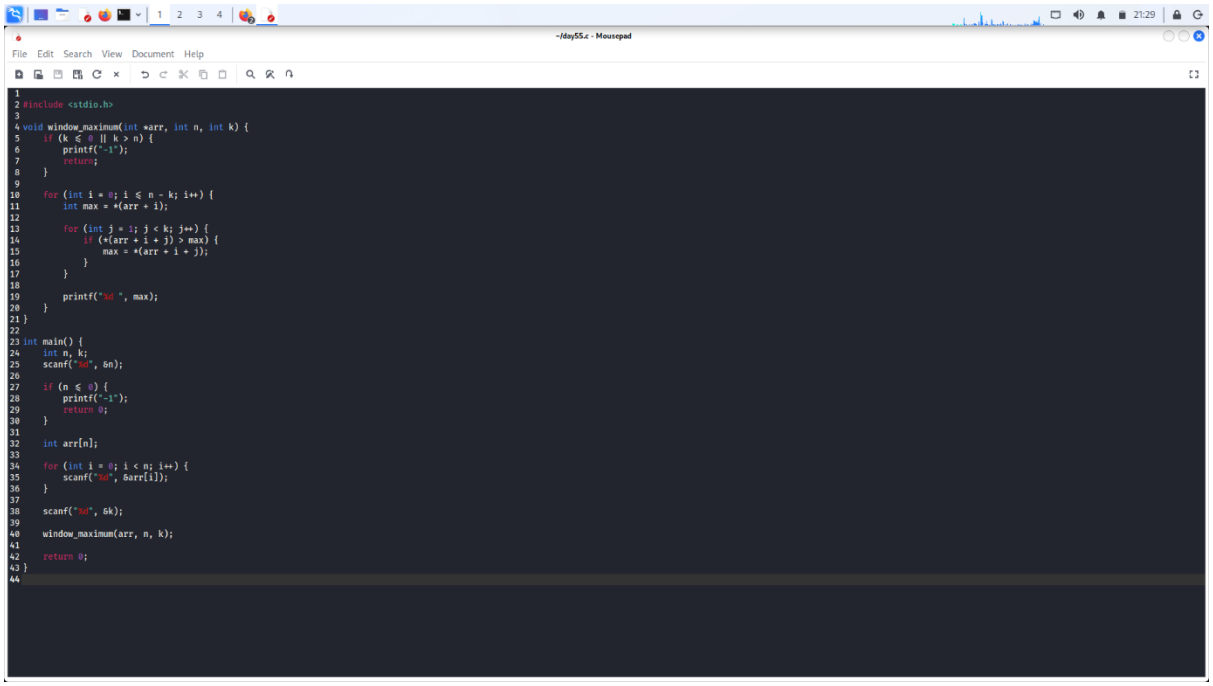
Aim

To find the maximum element in each sliding window of an integer array using pointers, functions, and nested loops in C.

Algorithm

1. Start.
2. Read the size n of the array.
3. Declare a variable-length array arr[n].
4. Read n array elements.
5. Read the window size k.
6. Pass the array to window_maximum() using its address.
7. If k <= 0 or k > n, print -1.
8. Otherwise, use an outer loop to move the window.
9. Use an inner loop to find the maximum element in each window.
10. Print the maximum elements separated by spaces.
11. Stop.

>C PROGRAM:



```c
#include <stdio.h>

void window_maximum(int *arr, int n, int k) {
    if (k <= 0 || k > n) {
        printf("-1");
        return;
    }

    for (int i = 0; i <= n - k; i++) {
        int max = *(arr + i);

        for (int j = 1; j < k; j++) {
            if (*(arr + i + j) > max) {
                max = *(arr + i + j);
            }
        }

        printf("%d ", max);
    }
}

int main() {
    int n, k;
    scanf("%d", &n);

    if (n <= 0) {
        printf("-1");
        return 0;
    }

    int arr[n];

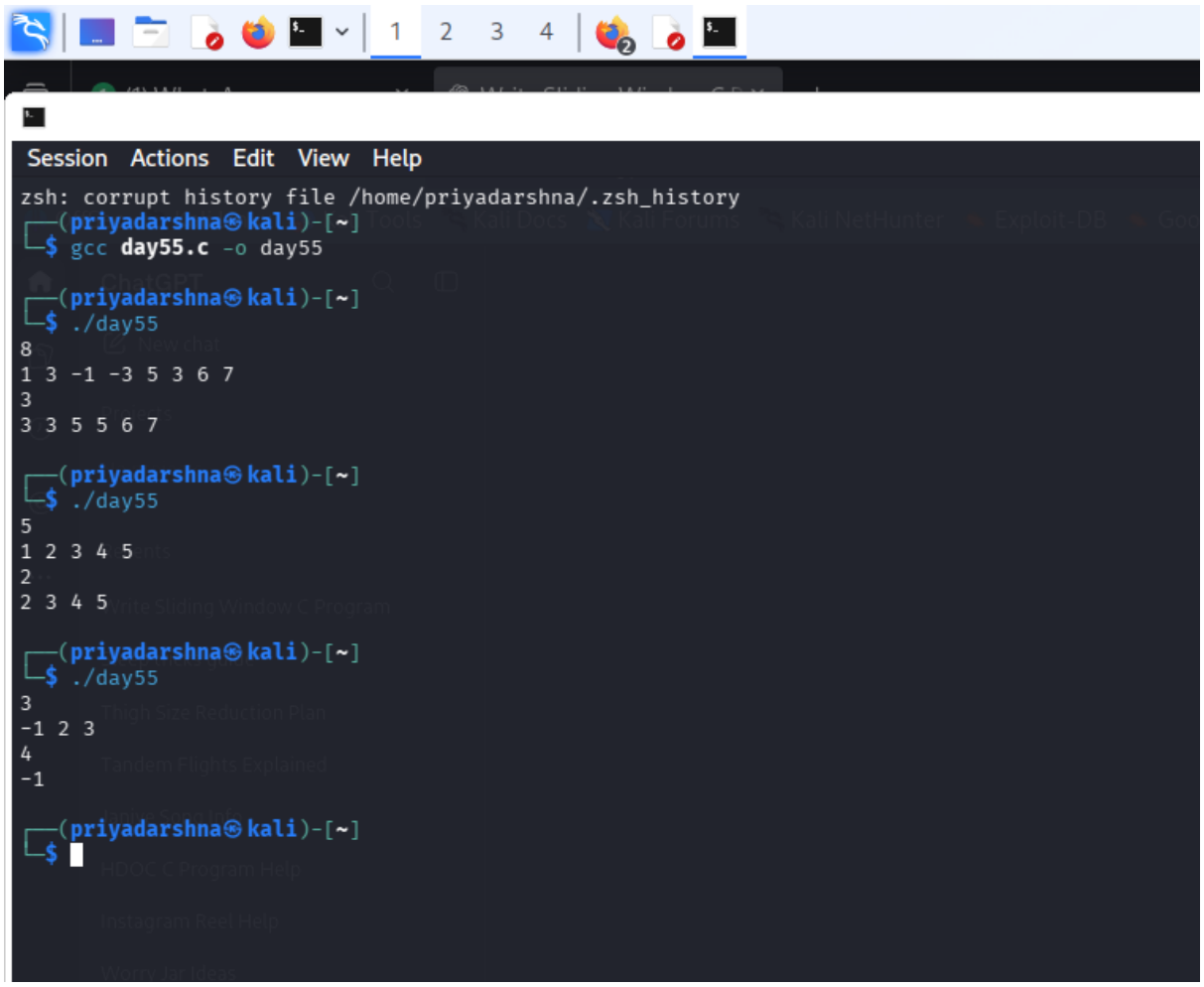
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &k);

    window_maximum(arr, n, k);

    return 0;
}
```

>OUTPUT:



```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day55.c -o day55

┌──(priyadarshna㉿kali)-[~]
└─$ ./day55
8
1 3 -1 -3 5 3 6 7
3
3 3 5 5 6 7

┌──(priyadarshna㉿kali)-[~]
└─$ ./day55
5
1 2 3 4 5
2
2 3 4 5

┌──(priyadarshna㉿kali)-[~]
└─$ ./day55
3
-1 2 3
4
-1

┌──(priyadarshna㉿kali)-[~]
└─$
```